



PTT PHENOL COMPANY
LIMITED

PROCESS SPECIFICATION

CRUDE PRODUCT COOLER (E-2309)

Docu. No. : 14780-8120-PS-E2309
Project No. : 120117
Rev. No. : Z1
Date : MAY 11, 2016
Page : 1 / 7

TITLE :

CRUDE PRODUCT COOLER

(E-2309)

DISCIPLINE : **PROCESS**
DOCUMENT NO. : **14780-8120-PS-E2309**
PROJECT NAME : **PTT Phenol Train II Project**
PROJECT NO. : **120117**
OWNER : **PTT Phenol Company Limited (PPCL)**

Z1	MAY 11, 2016	AS BUILT		S.W.Park	J.S.Chung	- S.W.Lee
F1	MAY 28, 2014	FINAL ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
A1	OCT 8, 2013	APPROVED FOR DESIGN		S.W.Park	J.S.Chung	- S.W.Lee
O1	JUN 27, 2013	COMMENT ISSUE		S.W.Park	J.S.Chung	- S.W.Lee
REV. NO.	DATE	DESCRIPTION		PREP'N	REVIEW	QA REVIEW APPROVAL
REV. NO.	PREPARATION DEPT.	PREPARATION	REVIEW	QA REVIEW		APPROVAL
Z1	PROCESS	S.W.Park	J.S.Chung	-		S.W.Lee
		MAY 11, 2016	MAY 11, 2016	-		MAY 11, 2016

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SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units

Metric

Doc. No. : 14780-8120-PS-E2309

Rev. No. : Z1

Item No. : E-2309

Page : 3 / 7

Rev

Case	Design	TEMA Type	AES	Orientation	Horizontal
Number of Units	1	Shells per Unit	1	Shells in Series	1
Surface per Unit m ²	227.0	Surface per Shell m ²	227.0	Shells in Parallel	1

PERFORMANCE OF ONE UNIT

	In	SHELLSIDE	Out	In	TUBESIDE	Out
Fluid Name		Cooling Water			Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O	
Total Flowrate (Note 2) kg/h		338569 × 1.05			61258 × 1.05	
Vapour kg/h		0	0		0	0
Liquid kg/h		338569 × 1.05	338569 × 1.05		61258 × 1.05	61258 × 1.05
Fluid Condensed kg/h		0			0	
Fluid Evaporated kg/h		0			0	

LIQUID PROPERTIES

	AS WATER	SEE PAGE 7
Density kg/m ³		
Viscosity cP		
Specific Heat kcal/kg.°C		
Thermal Conductivity kcal/h.m.°C		
Surface Tension dyne/cm		
Pour Point Temperature °C		

VAPOUR PROPERTIES

Density kg/m ³		
Viscosity cP		
Specific Heat kcal/kg.°C		
Thermal Conductivity kcal/h.m.°C		
Hydrate Temperature °C		

THERMAL AND HYDRAULIC DATA

Operating Temperature °C	33.0	43.0	145.0	43.0
Operating Pressure kg/cm ² g	3.867		5.083	
Number of Passes per Shell	1		8	
Average Velocity m/s	0.8		1.4	
Pressure Drop (Allow / Calc) kg/cm ²	0.703	0.345	0.703 (Note 10)	0.837
Fouling Factor h.m ² .°C/kcal	0.0004		0.0006	

Duty (Note 2) kcal/h × 10 ⁶	3.384 × 1.05	MTD °C	34.9
U-value Service/Clean kcal/h.m ² .°C	459	Over Surface %	2.5

DESIGN CONDITIONS

	SHELLSIDE	TUBESIDE
Design pressure max/min kg/cm ² g	16.5	21.0
Full Vacuum	Not Required	Required
Test Pressure kg/cm ² g	To Code	To Code
Design Temperature max/min °C	120	250
Corrosion Allowance mm	3.2	1.5
Insulation Details	None	Personnel Protection
Hydrogen Service	No	No
Sour Service	No	No
Lethal Service	No	No
Cyclic Service	No	No
Cleaning Method	Mechanical	Mechanical
Stress Relief for Process Reasons	No	No
Heat Release Curve	No	Yes

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Rev. No. : Z1

Item No. : E-2309

Page : 4 / 7

CONSTRUCTION OF ONE SHELL

SHELL	Shell I.D.	mm	860	Channel I.D. (If different)	mm	
TUBES	Number of Tubes		650	Tube Length	mm	6096
	Tube O.D.	mm	19.05	Tube Thickness	mm	1.651
	Tube Gauge	BWG	16	Tube Wall Basis		Average
	Tube Pitch	mm	25.4	Tube Layout Angle	deg.	90
	Tube-Tubesheet Joint		Expanded	Tube-Baffle Diam. Clearance	mm	0.8
BAFFLES	Number of Baffles		12	Baffle Type		Single Seg.
	Baffle Cut	%	33.8	Baffle Spacing	mm	420
	Baffle Cut Orientation		Horizontal	Double Seg. Baffle Overlap	mm	0
	Baffle Thick	mm	6.4	Baffle-Shell Diam. Clearance	mm	4.8
BUNDLE	Impingement Protection		Rods (Note 4)	Tube Supports (Note 3)		At Floating Head
	Pairs of Sealing Strips		3	Sealing Strip Details (Note 6)		Maximum Length
	Passlane Sealing		Yes	Passlane Sealing Details (Note 6)		Maximum Length
	U-Bend Diam.	mm	N.A.	U-Bend Orientation		N.A.
	U-Bend Area Included		N.A.	U-Bend Support Details		N.A.
	Longitudinal baffle		N.A.	Longitudinal Baffle Seal		N.A.

CONNECTIONS

SHELLSIDE						TUBESIDE					
Service	No.	N.B.		Rating	Facing	Service	No.	N.B.		Rating	Facing
Inlet S1	1	10	Inch	300	RFWN	Inlet T1	1	6	Inch	300	RFWN
Outlet S2	1	10	Inch	300	RFWN	Outlet T2	1	6	Inch	300	RFWN
Vent SV	1	2	Inch	150	RFLWN	Vent TV	1	3/4	Inch	300	RFLWN
Drain SD	1	2	Inch	150	RFLWN	Drain TD	1	3/4	Inch	300	RFLWN

MATERIALS

Tubes	304L Stainless Steel (Seamless)
Shell / Shell Cover	Killed Carbon Steel (UOP Std 3-11)
Floating Head Cover	304L Stainless Steel (UOP Std 3-15)
Channel / Channel Cover	304L Stainless Steel (UOP Std 3-15)
Tubesheet, Stationary	304L Stainless Steel
Tubesheet, Floating	304L Stainless Steel
Baffles / Tube Supports	304 Stainless Steel
Nozzles / Nozzle Flange	Shellside: Killed Carbon Steel
Nozzles / Nozzle Flange	Tubeside: 304L Stainless Steel
Bolt and nuts	Per Project Specifications
Gaskets, Shellside	Per Project Specifications
Gaskets, Tubeside	Per Project Specifications
Gaskets, Nozzles	Per Project Specifications

ESTIMATED WEIGHTS

Weights per Shell	Exchanger Weight, Dry	8.3	Exchanger Weight, Wet	12.1
(Tonnes)	Bundle Weight, Dry	3.6		

DESIGN REQUIREMENTS

PED Classification	Shellside- N.A.	Tubeside- N.A.
Project Specifications	UOP 963766 - 401, UOP Std 4-11-7, UOP Std 3-11, UOP Std 3-15, UOP Std 9-11-7, 14780-21A1	
Design Code	ASME VIII Div 1	
Equipment Code	TEMA Class R 9th edition, API 660 8th edition 2007	

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Rev. No. : Z1

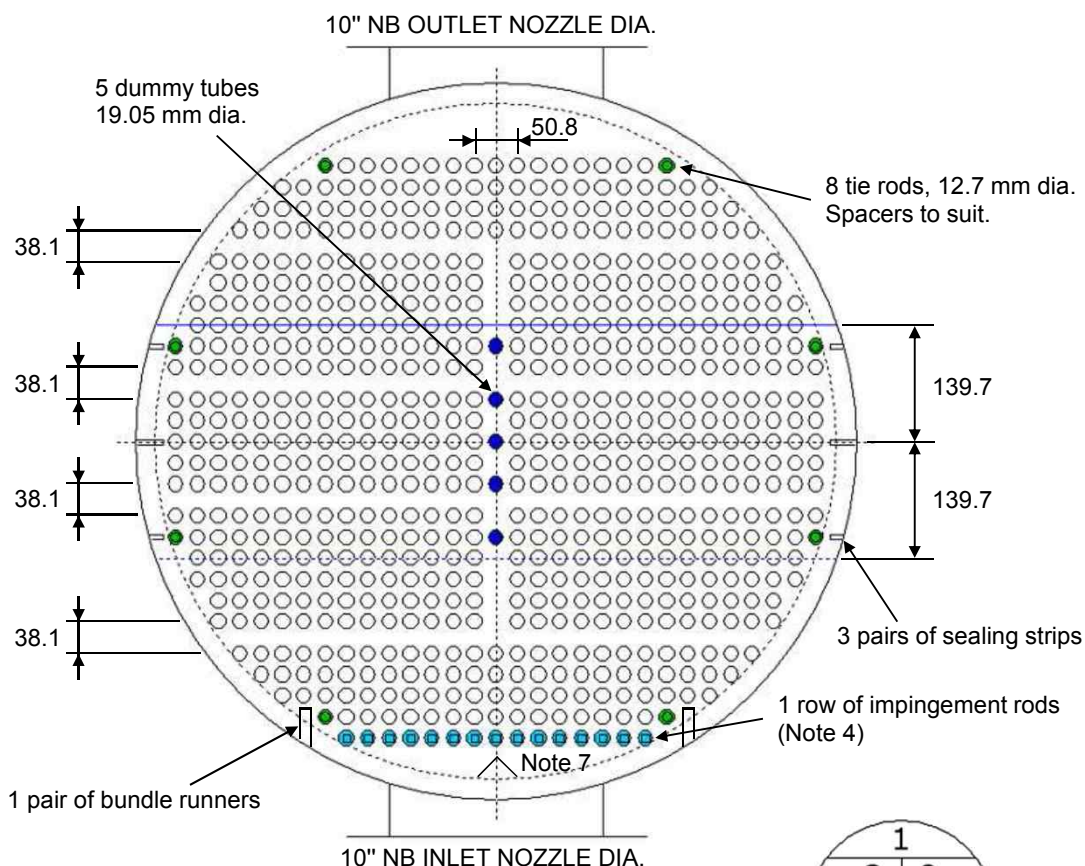
Item No. : E-2309

Page : 5 / 7

Notes:

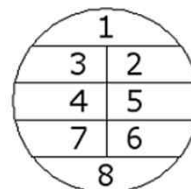
1. Item has been checked for Flow Induced Vibration. Tube damage is not predicted.
2. Flow and duty include 5% design margin.
3. Provide a floating head support plate.
4. Provide 1 row of impingement rods of the same diameter, same pitch, and same layout as the tubes.
5. Bundle runners shall not restrict the shellside entrance flow path.
6. Sealing strips, bundle runners, and dummy tubes shall not restrict the bundle cleaning lanes.
7. All baffles and support plates extending to the bottom of the shell shall be provided with V-notch at the lowest point.
8. A Magnesium anode shall be provided on the shellside.
9. Test ring shall be provided by VENDOR.
10. The allowable tubeside pressure drop to be confirmed later.
11. Deleted.

Tube Layout



O.T.L. = 812.6 mm

All dimension in mm
Sketch not to scale



Tubepass Arrangement

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

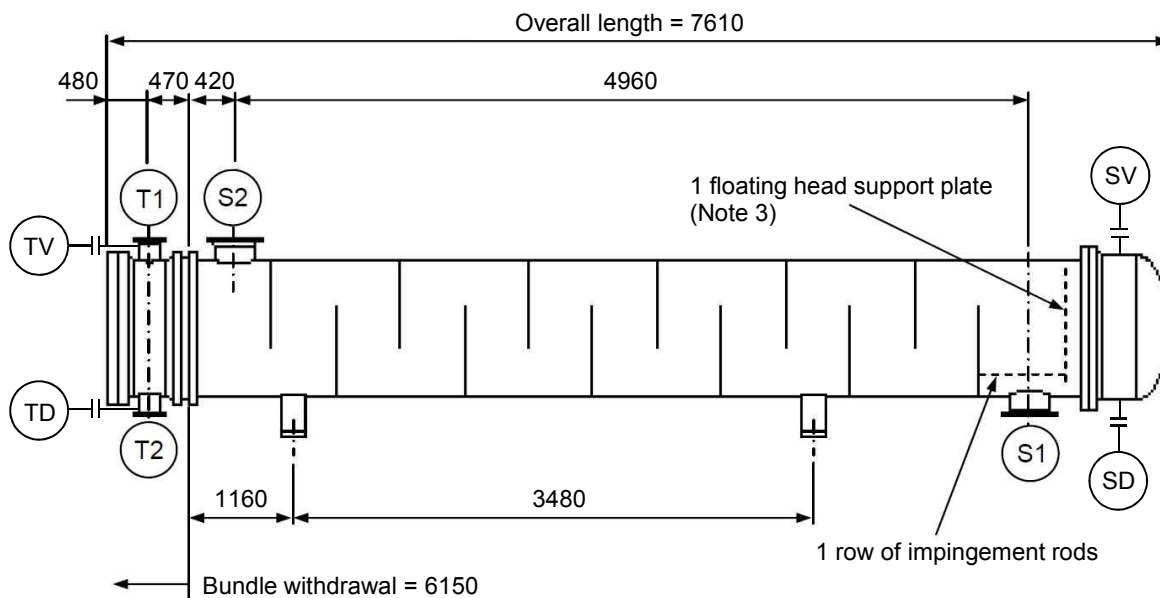
Doc. No. : 14780-8120-PS-E2309

Rev. No. : Z1

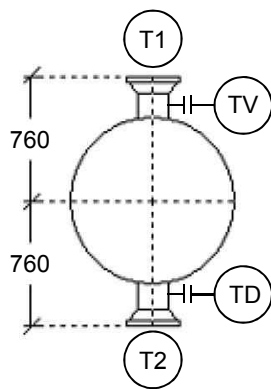
Item No. : E-2309

Page : 6 / 7

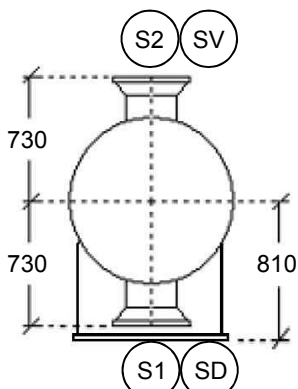
Setting Plan



Tubeside Nozzles



Shellside Nozzles



Shell ID : 860 mm

Tube length : 6096 mm

All dimension in mm.

Drawing not to scale

PROCESS NOZZLE SUMMARY

REF	SERVICE	SIZE	RATING	ELEV.	REMARK
S1	Cooling Water Inlet	10"	300#RFLWN		
S2	Cooling Water Outlet	10"	300#RFLWN		
T1	Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O Inlet	6"	300#RFLWN		
T2	Phenol, Acetone, Cumene, H ₂ SO ₄ , H ₂ O Outlet	6"	300#RFLWN		
SV	Shellside Vent	2"	150#RFLWN		CW/blind
SD	Shellside Drain	2"	150#RFLWN		CW/blind
TV	Tubeside Vent	3/4"	300#RFLWN		CW/blind
TD	Tubeside Drain	3/4"	300#RFLWN		CW/blind

Rev

SHELL AND TUBE EXCHANGER THERMAL DATA SHEET

System of Units Metric

Doc. No. : 14780-8120-PS-E2309

Rev. No. : Z1

Item No. : E-2309

Page : 7 / 7

HotSide Property Curve

Pressure

5.083

kg/cm²g

Temperature °C	Entire Stream Enthalpy kcal/kg	Entire Stream Mass Vapour fraction	Vapour Density kg/m ³	Vapour Viscosity cP	Vapour Therm. Cond. kcal/h.m.°C	Vapour Spec. Heat kcal/kg.°C
145.0	55.20	0.0000	-	-	-	-
134.9	49.06	0.0000	-	-	-	-
124.6	42.93	0.0000	-	-	-	-
113.9	36.80	0.0000	-	-	-	-
103.0	30.66	0.0000	-	-	-	-
91.7	24.53	0.0000	-	-	-	-
80.1	18.40	0.0000	-	-	-	-
68.1	12.27	0.0000	-	-	-	-
55.8	6.13	0.0000	-	-	-	-
43.0	0.00	0.0000	-	-	-	-

Temperature °C	Liquid Density kg/m ³	Liquid Viscosity cP	Liquid Therm. Cond. kcal/h.m.°C	Liquid Spec. Heat kcal/kg.°C	Liquid Surf. Tension dyne/cm	Critical Pressure kg/cm ² [a]	Critical Temperature °C
145.0	816.3	0.284	0.106	0.62	19.4	60.2	334
134.9	828.3	0.306	0.108	0.60	20.5	60.2	334
124.6	840.3	0.334	0.111	0.58	21.7	60.2	334
113.9	852.5	0.368	0.113	0.57	22.9	60.2	334
103.0	864.7	0.412	0.115	0.55	24.2	60.2	334
91.7	877.0	0.468	0.118	0.54	25.5	60.2	334
80.1	889.4	0.543	0.120	0.52	26.8	60.2	334
68.1	902.0	0.644	0.123	0.50	28.2	60.2	334
55.8	914.7	0.786	0.125	0.49	29.7	60.2	334
43.0	927.5	0.991	0.128	0.47	31.2	60.2	334

Rev